STEVENS CPE · KYRON COURSE DRAFT LAUNCHER

Course video automation: harness status, Kyron MCP read and write, test plan and cost evaluation

A checkpoint report before the team starts gathering testing sets. Updated after the proposed harness was built and QA made its first live fix in Kyron.

Prepared	November 2, 2026 (updated 1:30 PM ET)
Environment	Local development (`localhost:3000`), shared Neon database, OpenAI `gpt-6-luna`
Data window	Logged activity from Oct 30, 2026 8:16 PM to Nov 2, 2026 1:21 PM ET (log system enabled at the start of this window)
Kyron MCP	Connected Nov 2, 10:50 AM ET with read permission; reconnected 1:12 PM ET with **read write**
Screenshots	Captured from the running app on Nov 2, 2026 (admin view)

Summary

$0.030
Total model spend logged (29 calls, 0 failed)

8
QA runs: 4 setup passed, 4 lesson (2 read from Kyron MCP)

1
Live QA fix written to Kyron: lock version 2 → 3, verified

103
Kyron MCP tools; draft 14826 read and written (7 activities, 18 scenes)

- **The proposed harness is built** (section 2). One production job per course, advanced by a scheduler that also polls Kyron, so nothing depends on an open page. Lessons are read from Kyron MCP scene by scene. Fixes are written back to the exact scene with Kyron's edit lock. An independent verifier can undo a bad fix. A video render and narration loop is built but off by default. A defect library collects problems for the testing sets.
- **First live write (section 3):** one false sentence was planted in a scene of the "[QA Test]" draft. QA found it as critical, proposed a one-sentence fix and escalated it. After a person approved, QA wrote it to Kyron, read it back, the verifier confirmed it, and the re-check passed. Total model cost: $0.009.
- **Cost:** a full check of the real 7-activity lesson costs about $0.004, or $0.009 with a fix, verification and re-check. The test plan in section 4 stays at about $5 in model costs ($10 with contingency). People's review time (about 28 hours) and Kyron credits are the real costs.

Contents

1. Kyron MCP dataset: what can be pulled
2. Current harness setup (as built)
3. Live write test on draft 14826
4. Testing plan: verify the harness and gather problems
5. Cost evaluation
6. Appendix: records referenced

1. Kyron MCP dataset: what can be pulled

> **Connected on November 2, 2026 at 10:50 AM ET with read permission, and at 1:12 PM ET with read write.** The app works with Kyron through **103 MCP tools** (39 read-only). Draft 14826 was read end to end: collection → lesson → section → 7 activities → 18 scenes. The data in 1.3 and 1.4 was captured before any write. Everything pulled, with media links removed, is saved in `reports/kyron-mcp-dataset-draft-14826-2026-11-02.json` for the testing-set phase.

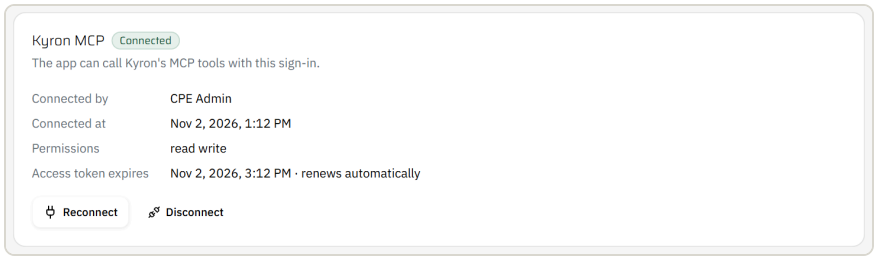


Figure 1. Kyron connection (admin): connected with *read write* permission; the sign-in renews automatically.

1.1 How the connection works

- **Client file.** Kyron identifies the app by a public client file (a client ID metadata document). The file is hosted in a secret GitHub gist and served through a JSON mirror (githack). It holds no secrets: only the app name and the address Kyron returns to after sign-in.
- **Lesson learned.** Kyron rejected the file while GitHub served it as plain text (`text/plain`), showing "Client authentication failed due to unknown client…". It accepted the same file once served as `application/json`.
- **Sign-in.** One Kyron account signed in once. The tokens are stored encrypted on this computer and renew automatically (access token lifetime 2 hours).
- **Permission.** `read write`. Two switches limit what QA may change. `QA_KYRON_WRITES=test-only` allows writes only to drafts named "[QA Test] …". `QA_KYRON_VIDEO=off` means no renders, since they use Kyron credits. QA never publishes, deletes scenes, regenerates scenes or changes themes and avatars.
- **Before wider use,** move the client file to a host the team controls, such as GitHub Pages or the deployed app, instead of a third-party mirror.

1.2 What Kyron MCP exposes

Area	Read	Write	Tools the harness uses (bold) and other notable tools
Lesson content	9	15	**get_lesson, list_lesson_sections, list_lesson_activities, get_lesson_activity, list_vocabulary_items, validate_lesson, get_lesson_lock_version, update_lesson_activity, update_lesson, update_lesson_section, update_vocabulary_item**; section generation
Video and scenes	8	4	**regenerate_all_lesson_videos**, get_lesson_segment, get_lesson_segment_transcript; avatars, layout previews
Lesson collections (drafts) and versions	5	11	**get_lesson_collection**, list_lesson_collection_versions, copy_lesson_collection; publish_lesson_collection (never automatic)
AI rewrite (regeneration)	1	1	**create_regeneration_log, get_regeneration_log**: Kyron's own AI rewrites one field
Learner runs	4	5	lesson instances, transcripts, misconceptions found in a learner's run (for tutor checks)
Library folders and files	2	10	folders, uploads, attachments
Comments	2	6	comment threads on activities: a possible channel for escalations
Organization, access and keys	5	8	not used (access policies, API keys, organizations)
Integrations (Canvas, LTI)	3	4	not used
Total	**39**	**64**	**103 tools; the app's explorer runs read-only tools only**

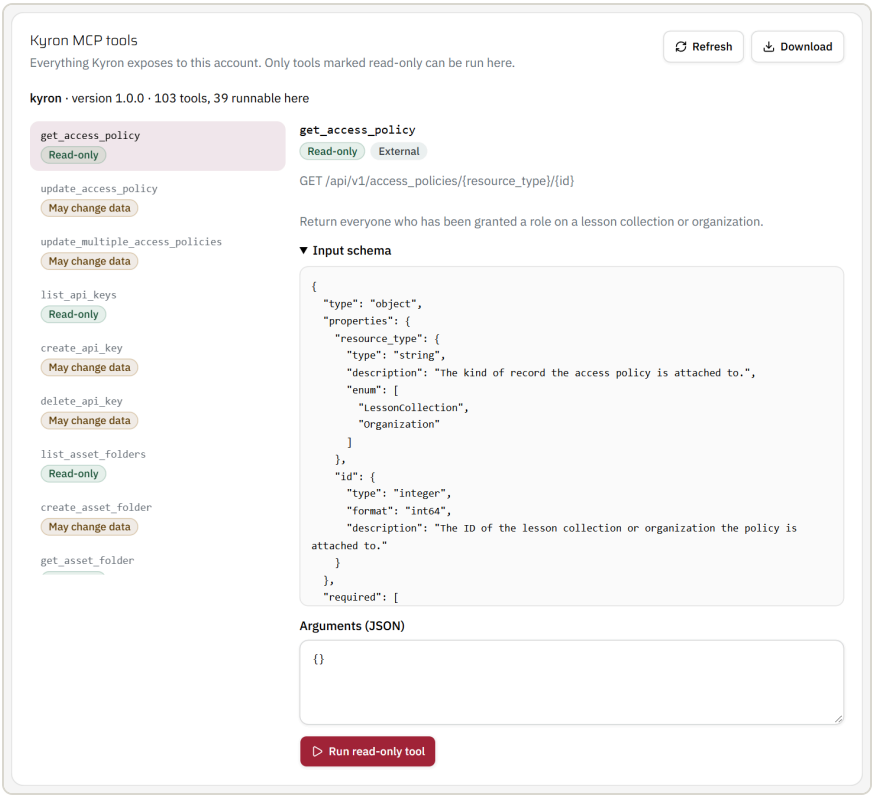


Figure 2. The tool explorer ("MCP terminal") in the app: every Kyron tool with its access level; only read-only tools can be run.

1.3 Data pulled for draft 14826

Level	What came back
Collection 14826	"[QA Test] Introduction to PDEs - MIT 18.152 Class 1" · type `one_shot_course` · status done · not published · description
Draft lesson 23930	Audience, objective, intended time 900 s, avatar "Marcus D", locale en, edit lock version 1, theme and branding, **8 learner misconceptions**, **8 vocabulary terms**
Section 64011	"PDE Order and Linearity": 1 structured objective, 1 enduring understanding, 3 essential questions, knowledge and misconceptions
Activities (7)	Each with type, title, order and branching, plus its full authored content (below)
Scenes (18)	Title, on-screen text, narration script, layout, image, caption file, rendered video address and status

Activity	Kyron type	Title	Scenes
504101	introduction	Introduction	2
504102	reflection_conversation	Think of a time when paying close attention to small details helped you…	2
504103	lecture	Read PDE Notation	4
504104	clarifying_conversation	In one sentence, restate the rule for determining a PDE's order…	2
504105	lecture	Linearity in PDEs	4
504106	clarifying_conversation	Why is the equation u_t = a(x)u_xx + b(x)u still linear…?	2
504107	conclusion	Conclusion	2

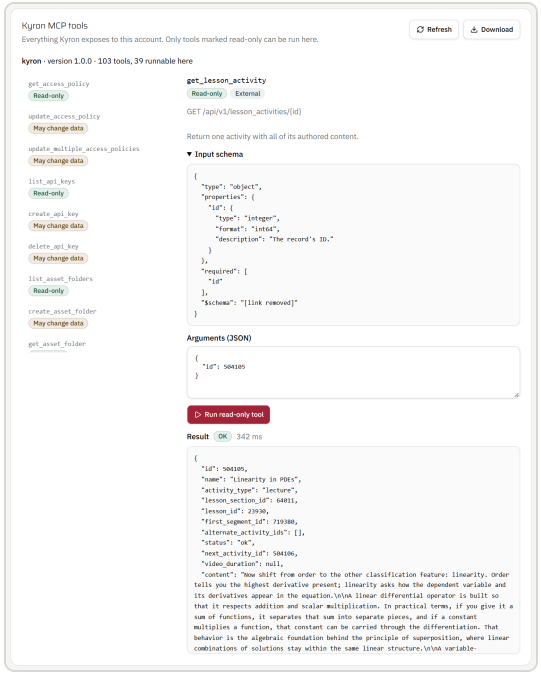


Figure 3. Pulling a lecture in the tool explorer: `get_lesson_activity` with `{"id": 504105}` returns the full script, then its scenes (signed media links removed in this screenshot).

1.4 What one lecture and one conversation contain

Lecture "Linearity in PDEs" (504105): 4 scenes, 242 narration words, captions for every scene, no rendered video yet.

#	Scene title	Layout	Narration (start)	On-screen text
0	Linearity vs Order	full_bleed_image	"Now shift from order to the other classification feature: linearity..."	-
1	Operator Linearity	image_info	"A linear differential operator is built so that it respects addition and scalar multiplication..."	-
2	Variable-Coefficient PDEs	text_info	"A variable-coefficient linear PDE is one where the coefficient functions depend on the independent variables..."	Coefficients depend on x or t · They multiply the unknown...
3	Linear vs Nonlinear Terms	full_bleed_image	"Differentiation itself is linear: the derivative of a sum is the sum of the derivatives..."	-

Conversation (504106): the question learners discuss, a model answer, 1 answer option, the knowledge it targets with 4 misconceptions ("Confusing linearity and order", "Belief that variable coefficients make an equation nonlinear", ...), a bank of 4 alternative questions, and its own question and wait videos.



Figure 4. `get_lesson_collection` (14826): collection, draft lesson and its settings.



Figure 5. `list_lesson_activities`: the 7 activities in learner order.

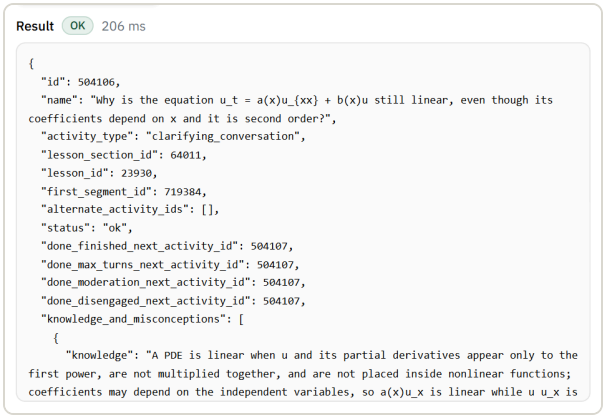


Figure 6. `get_lesson_activity` (504106): conversation question, answer and misconceptions.

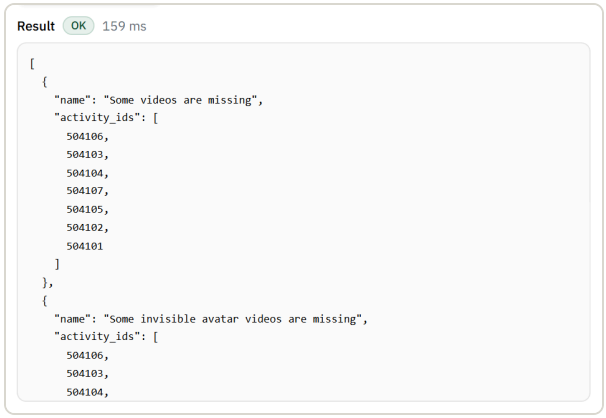


Figure 7. `validate_lesson` (23930): Kyron's own publish check reports missing videos on all 7 activities.

1.5 How Kyron's data maps to the QA harness

Harness snapshot field	Kyron MCP source	Tool
lesson.title, description, audience	Lesson name, description, audience	get_lesson
lesson.learningObjectives	Lesson objective + section structured objectives	get_lesson, list_lesson_sections
lesson.durationSeconds, assetType	intended_time, lesson_type	get_lesson
activity.id, type, title	Activity id, activity_type, name. Types: introduction, lecture and conclusion map to *instruction*; reflection_conversation to *reflection*; clarifying_conversation to *conversation*	list_lesson_activities
activity.narration, onScreenText	All scenes joined, for the checks (read-only)	get_lesson_activity
scenes.<id>.narration, title, onScreenText	Each scene's `scene_narration`, `scene_title`, `scene_text`: the fields fixes are written to	get_lesson_activity
activity.question	Conversations: question and model answer (an open question)	get_lesson_activity
vocabulary.<id>.term, definition	Vocabulary list, checked as a "Vocabulary" activity	list_vocabulary_items
activity.tutorGuidance	knowledge_and_misconceptions; lesson misconceptions	get_lesson_activity, get_lesson
Video and narration check	Scene caption files (VTT), scene video address, video_status	get_lesson_activity
References for fidelity	Kyron's `citations` field is empty for this lesson; the workspace sources stay the reference	-

Harness action (`write`)	Kyron tool and behavior	Status
Apply a fix to one scene or field	`update_lesson_activity` with `<part>.scenes[{id, scene_narration | scene_title | scene_text}]`, plus the lecture script `content` kept in step, and the lesson's `lock_version`. Kyron rejects a stale version, so QA never writes over a person's edit; the refusal is escalated	built, tested live
Fix a vocabulary entry or lesson setting	`update_vocabulary_item`; `update_lesson` and `update_lesson_section` (lesson settings always need a person's approval)	built
Render video once the script passes	`regenerate_all_lesson_videos`, then `validate_lesson` every 30 s; narration compared with the scene captions	built, off
Ask Kyron's own AI to rewrite a field	`create_regeneration_log`, then `get_regeneration_log`	not used
Publish	`publish_lesson_collection`: never automatic; needs a person's explicit confirmation	not built

1.6 Findings from the real Kyron data

#	Finding	What it means for the harness
1	**No video is rendered yet.** `validate_lesson` reports missing videos (and missing invisible-avatar videos) on all 7 activities; 0 of 18 scene videos exist.	Rendering is a separate step (`regenerate_all_lesson_videos`). That fits the plan: render only after the script passes QA.
2	**Segment and transcript reads are refused** ("Access forbidden"), with read permission and again with read write (retested Nov 2, 1:19 PM).	The narration check uses the scene caption files (15 of 18 scenes have them). Ask Kyron whether the account needs another role for segments.
3	**A claim not in the source:** the lecture and the vocabulary list cite the Black–Scholes equation. The MIT notes mention only the heat, wave and Laplace equations.	The fidelity check did not report it, because unsupported side examples are allowed by its rubric. Logged as the first defect (QA missed).
4	**Narration is short for the target time:** about 803 words (5–6 minutes) for a 15-minute lesson; conversations make up the rest.	Calibrate the duration check with real learner timings.
5	**No citations** are attached to any activity.	Grounding in the sources has to come from the harness's own fidelity check.

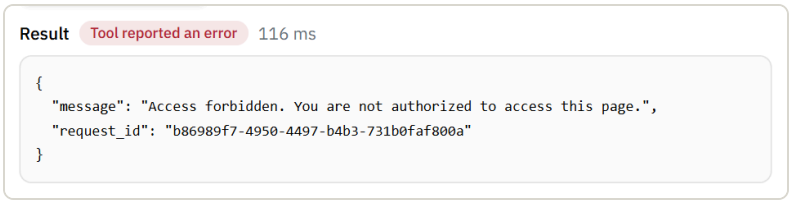


Figure 8. `get_lesson_segment_transcript`: refused by Kyron (the same with read write).

1.7 Before the connection: a lesson pasted into the harness

Before Kyron MCP was connected, lesson QA was tested on a short lesson in the snapshot format was pasted into the draft 14826 workspace. It was titled **"[QA Test] Seeded error - PDE classification"** and contained one deliberate false claim: that the heat equation is nonlinear. The checks found it as **critical** in both the narration and the on-screen text (see the workspace panel in section 2.2), and the revision agent proposed a correction. Pasted content cannot be changed by the agent, so the correction was stored as a downloadable proposed version.

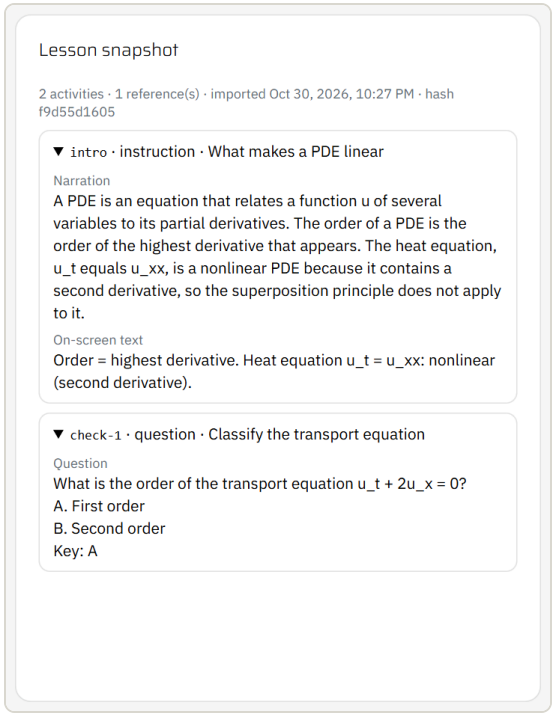


Figure 9. The snapshot as the checks see it: activities, narration, on-screen text, question and key.

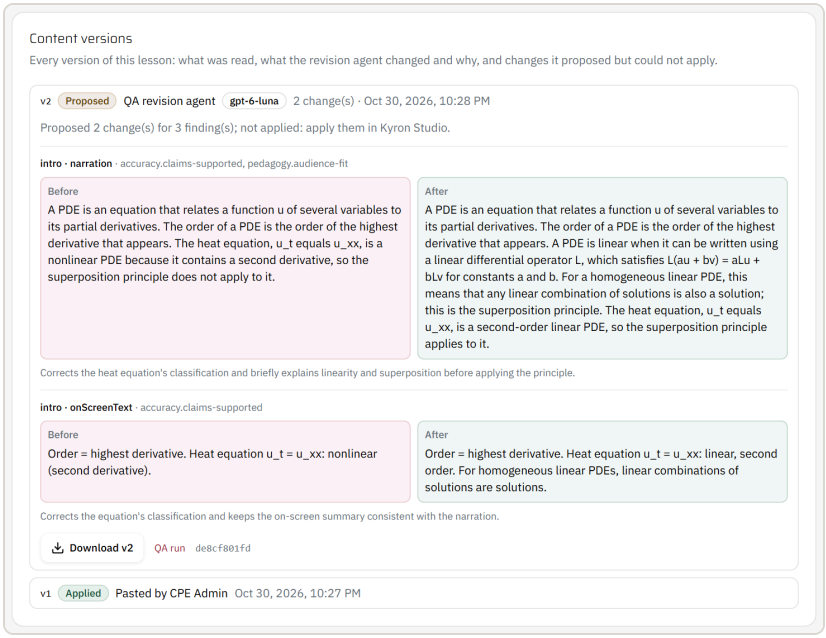


Figure 10. Content versions for the lesson: v1 as pasted, v2 proposed by the revision agent, with before and after for each field.

2. Current harness setup (as built)

The structure proposed in the first version of this report is now built. Each course is one durable **production job**, advanced by a scheduler tick: `npm run qa:worker` today, Vercel Cron when hosted. Steps are stored in Neon with leases, so a step that fails or times out is retried without running twice. Quality checks run **inside** generation. The generated lesson is read from Kyron through MCP, and fixes are written back there.

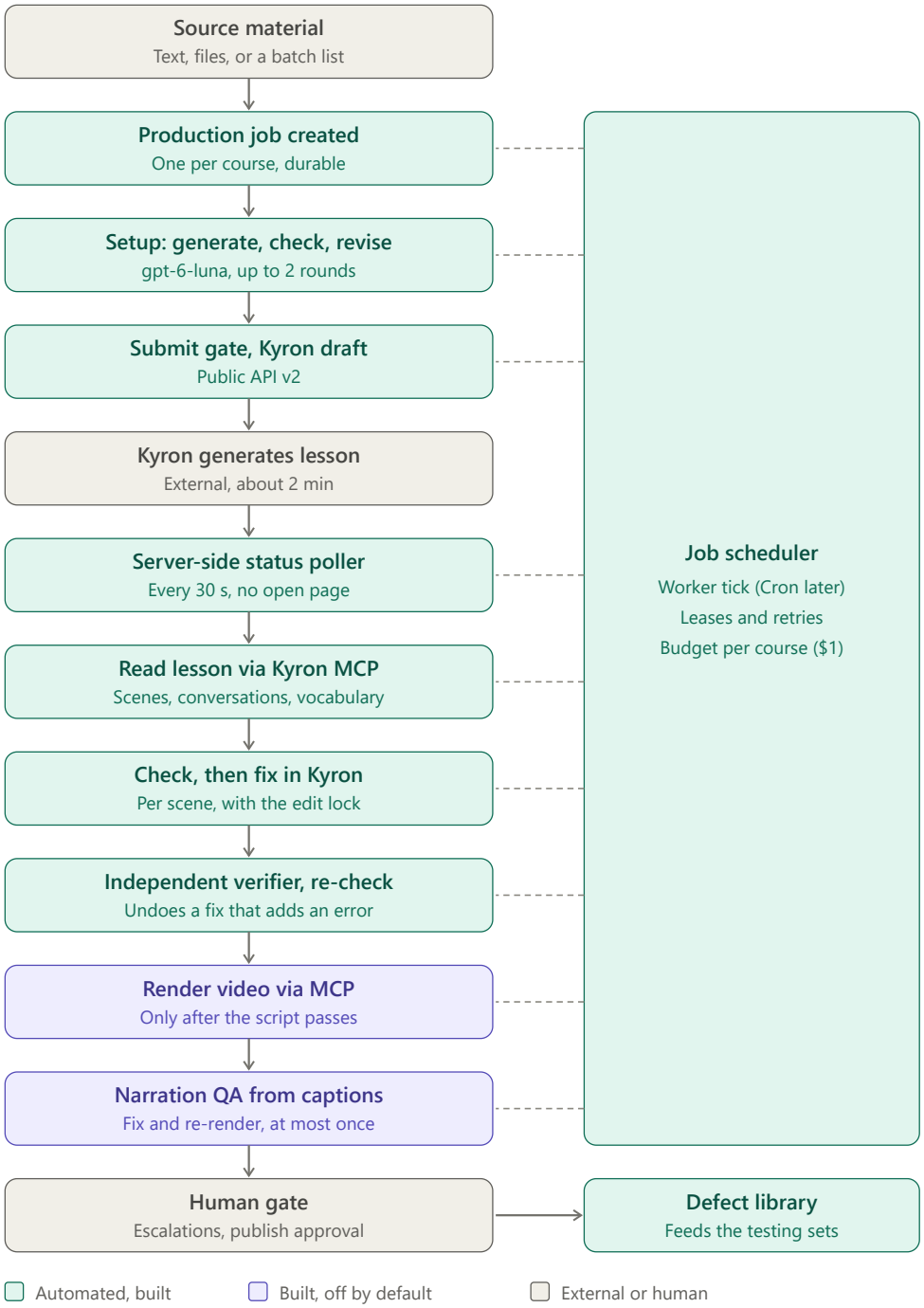


Figure 11. Current harness. After the Kyron handoff, everything runs without an open page. Fixes are written to the exact scene in Kyron. Critical fixes, answer keys and lesson settings wait for a person's approval. Video rendering is built but stays off until the team approves the Kyron credits.

2.1 Components

Stage	How it works	Status
Sign-in and roles	Preset access codes (admin, developer), signed session cookie, localhost only. Clerk kept for later hosting.	done
Production jobs and scheduler new	One job per course (single drafts and batches), with stage, what a person needs to do, and cost. Each tick polls Kyron, advances QA runs and updates every job. A run waiting on Kyron keeps a timed lease, so the worker does not spin. **QA agent → Production** lists them.	done
Course setup generation	gpt-6-luna (medium effort) builds name, description, objective, audience, duration and type from the sources.	done
Setup QA	Five criteria (source fidelity, measurable objective, grounded audience, duration and type fit, clear title), strict JSON output. Quoted evidence is verified against the sources.	done
Revision agent	One gpt-6-luna call per round fixes all open blocking findings together. The server checks each change, applies the safe ones (reading before and after the write), and keeps critical, answer-key and lesson-setting changes for approval. At most 2 rounds.	done
Independent verifier new	A separate gpt-6-luna call (high effort, different instructions) judges each applied fix: resolves, partial, does not resolve, or introduces an error. A fix that introduces an error is written back to its previous value and escalated.	done
Content versions	Every state of the setup or lesson is numbered, with author, model, reason, and before and after per field (per scene for Kyron lessons). Changes made in Kyron by anyone else are recorded too.	done
Submit gate and Kyron draft	Draft creation (public API v2, optional PDF/PPTX attachment) waits for a current, passing setup check. "Submit anyway" is recorded.	done
Status sync changed	Server-side poller in each scheduler tick: every generating draft, for every owner, at most every 30 s. A finished draft queues the lesson check.	done
Kyron MCP read and write new	Reads the lesson down to each scene, conversation and vocabulary term. Writes one scene or field at a time with the edit lock. Writes are limited to "[QA Test]" drafts (`QA_KYRON_WRITES=test-only`) until the harness is validated.	tested live
Lesson QA	Structure, blind assessment, source fidelity, consistency, alignment and audience checks, then the revision agent and verifier. Pasting a lesson still works without Kyron MCP.	done
Video and narration QA new	After the script passes: render through MCP, check status every 30 s, compare the scene captions with the script, fix the narration and re-render at most once. Uploading a video file also works.	built, off
Human checks and audit	Escalation queue, accept/reject/waive with a reason, tutor and visual checks, 10% random audit.	done
Defect library new	**QA agent → Defects**: reviewers log what QA missed, false alarms, wrong fixes, harness bugs, Kyron errors and source problems. Rejected findings and failed audits are added automatically. CSV/JSON export.	done
Logs and cost	Every model call (tokens, cost, latency), Kyron API and MCP calls (reads and writes), QA steps, versions and errors. Secrets are removed. CSV/JSON export.	done

2.2 The harness in the app

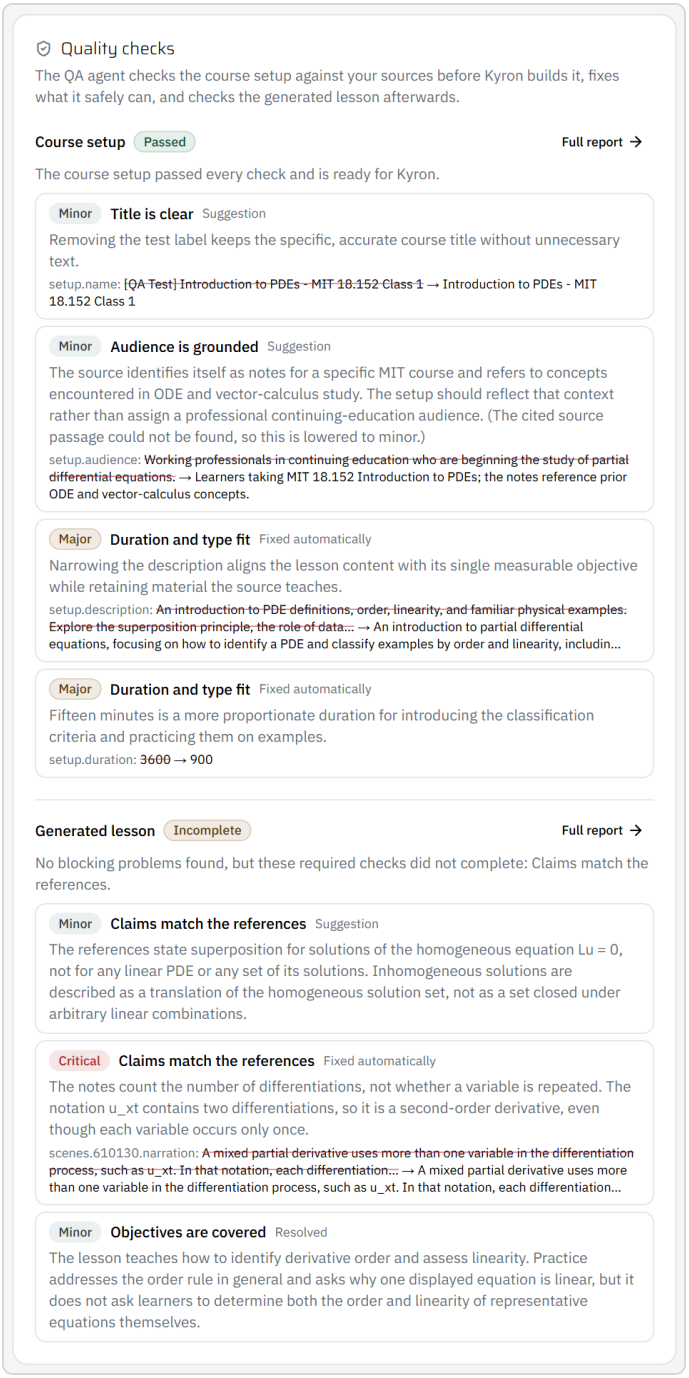


Figure 12. Quality checks on the workspace page of draft 14826. The course setup passed, with automatic fixes and two suggestions. The generated lesson was read from Kyron: the planted error was fixed after approval, and one minor finding is left for review. Two wording problems are visible: the approved fix is labeled "Fixed automatically", and the shortened before/after text hides where the sentence changed.

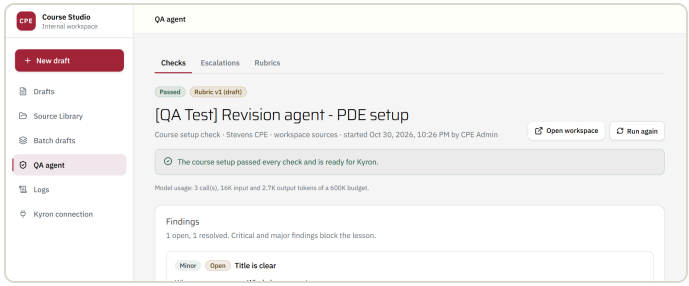


Figure 13. A setup QA run that passed after the revision agent's fix.

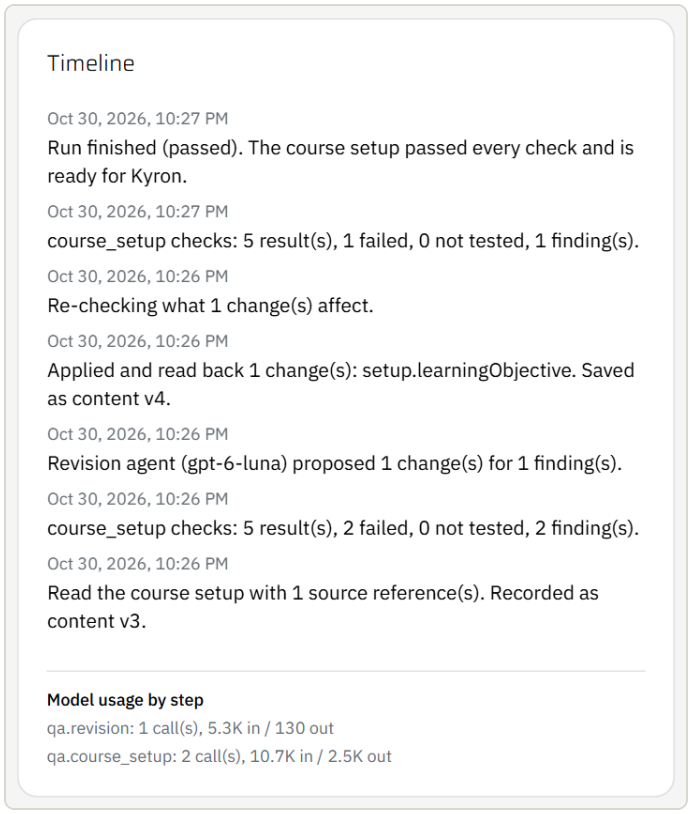


Figure 14. Timeline for that run: read, recorded as v3, checks, revision agent (gpt-6-luna), applied and read back as v4, re-check, passed.

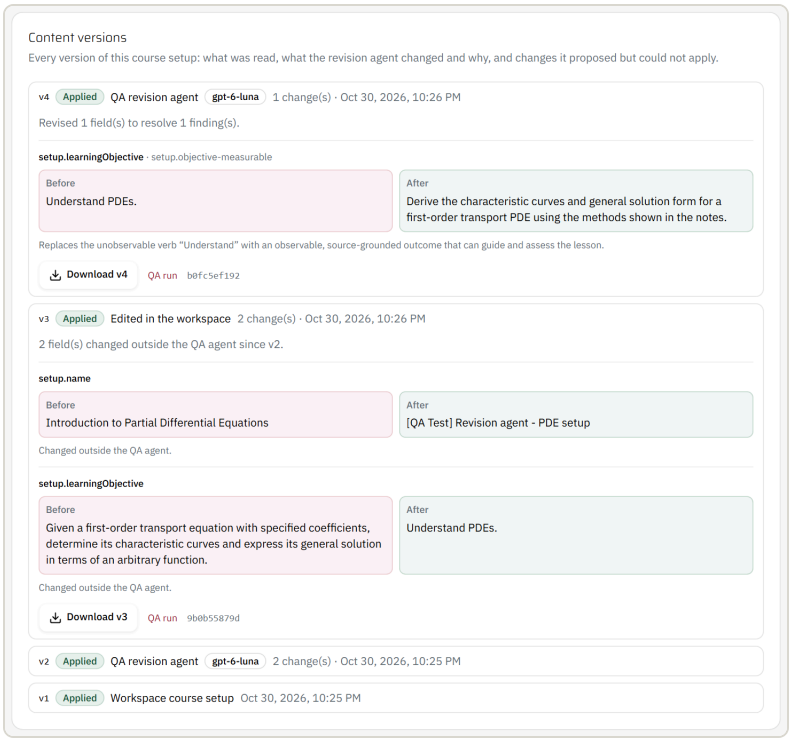


Figure 15. Content versions for a course setup. v3 records a person's edit (the vague objective "Understand PDEs."). v4 is the revision agent's fix, with the finding it resolves and its reason.

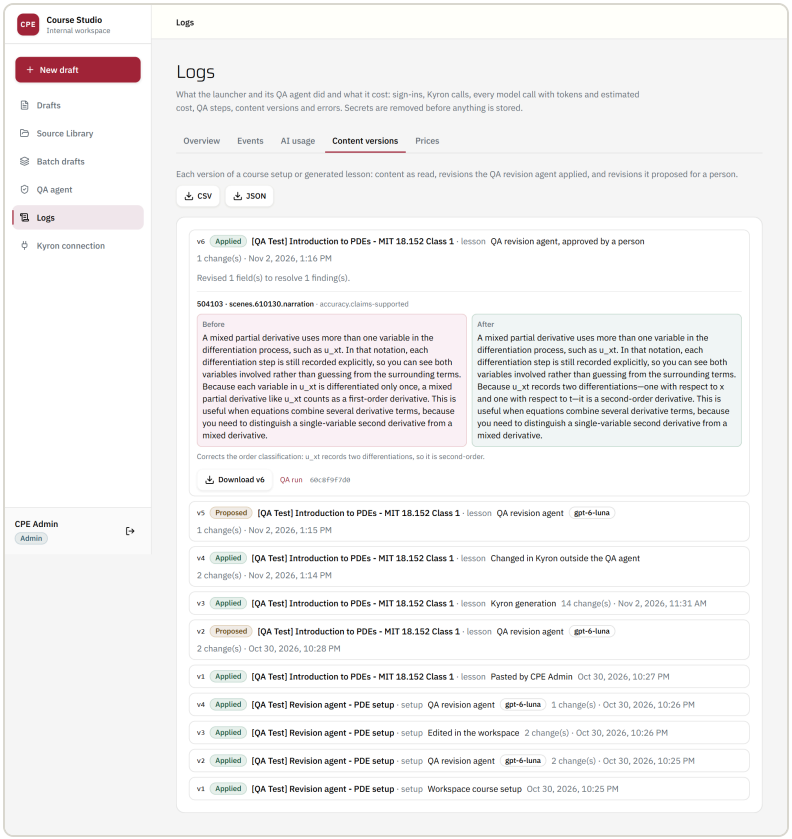


Figure 16. **Logs → Content versions**: every version across workspaces, with CSV and JSON export (one row per changed field). The newest is QA's fix to scene 610130.

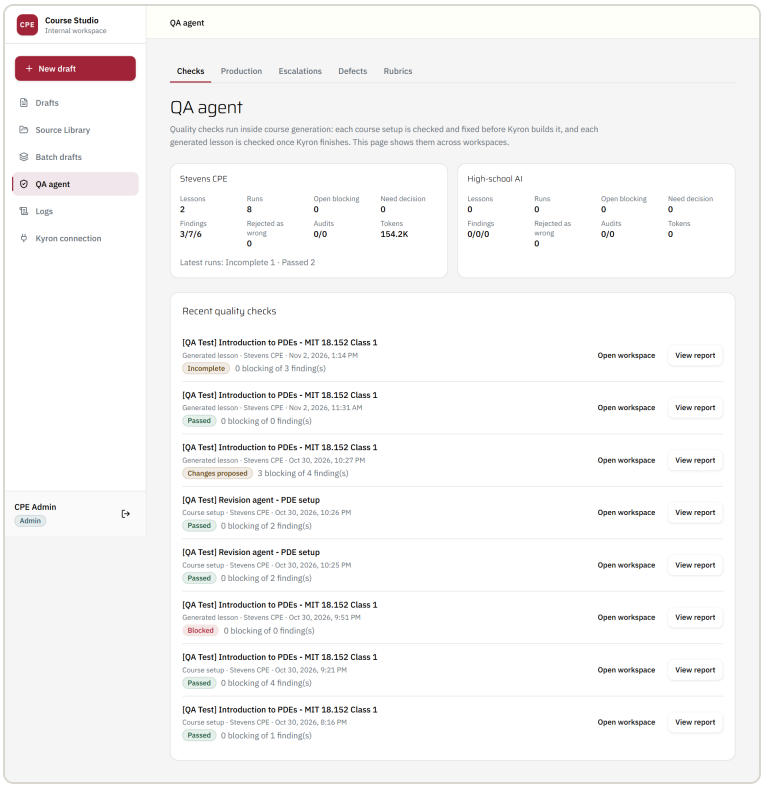


Figure 17. QA agent dashboard: every quality check across workspaces. The test content is named with a "[QA Test]" prefix.

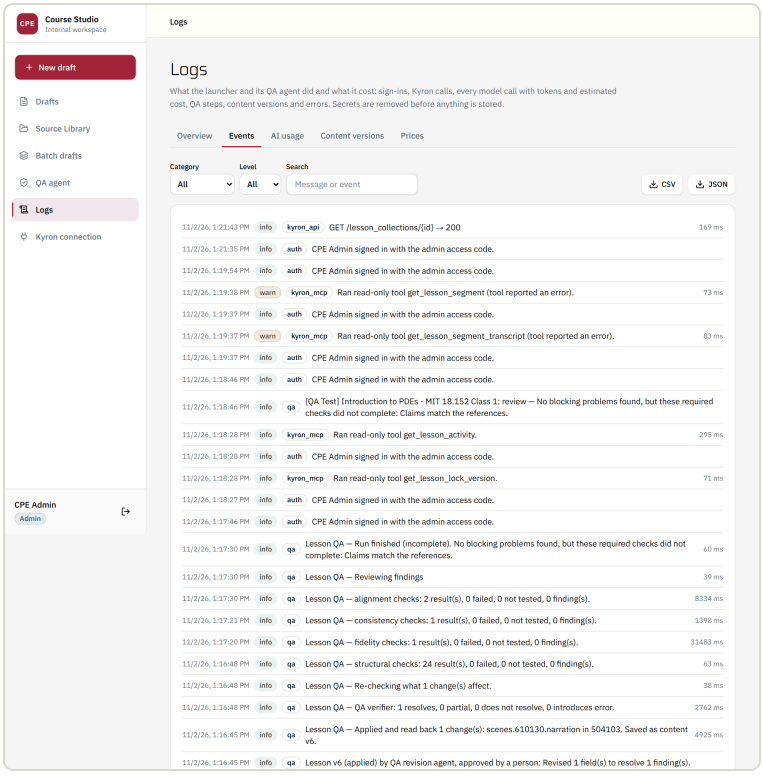


Figure 18. **Logs** → **Events** during the write test: QA's reads and writes in Kyron, verifier and re-check steps, content versions and sign-ins (the sign-ins are the test scripts). Secrets are removed.

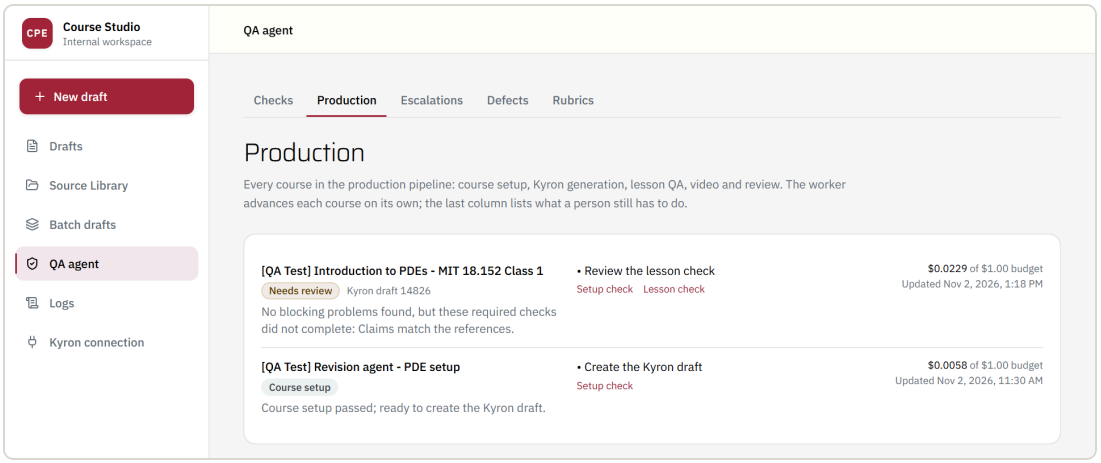


Figure 19. **QA agent → Production**: one row per course with its stage, what a person still has to do, links to its checks, and model spend against the course budget.

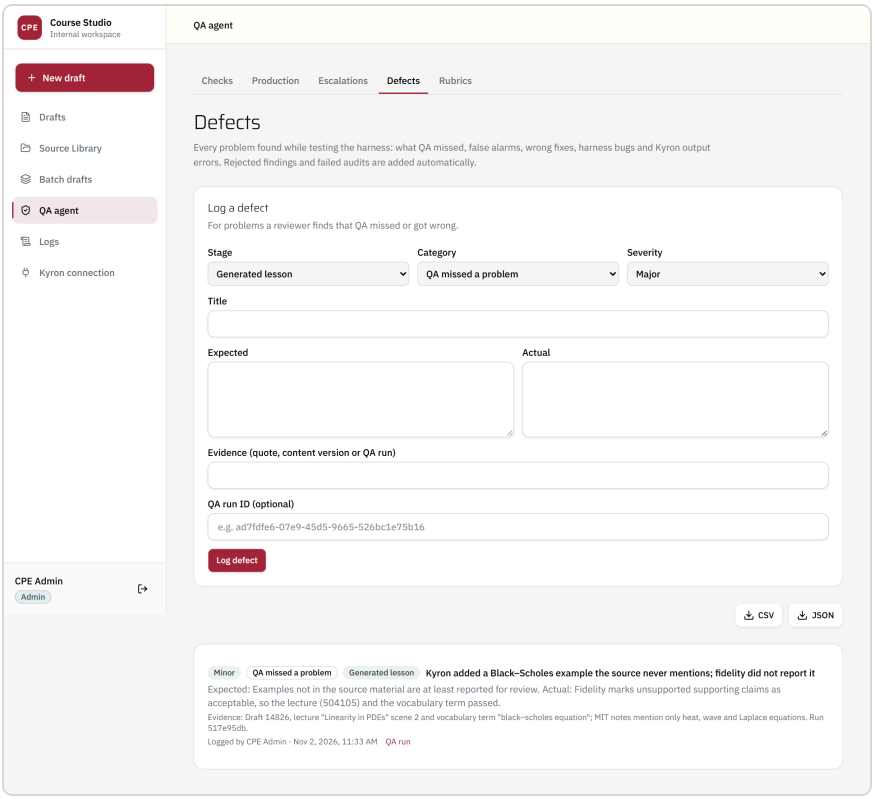


Figure 20. **QA agent → Defects**: the defect library. The first entry is the Black–Scholes example that the fidelity check did not report.

3. Live write test on draft 14826

To test the write path end to end, one false sentence was planted in a lecture scene of the "[QA Test]" draft using the harness's own Kyron writer. The lesson check was then re-run the same way it runs after Kyron generation. The planted sentence went into scene 610130 of lecture "Read PDE Notation" (activity 504103):

> "Because each variable in u_xt is differentiated only once, a mixed partial derivative like u_xt counts as a first-order derivative." This is false: u_xt records two differentiations, so it is second order. The MIT notes define the order as the maximum number of derivatives.

3.1 What happened

Time (ET)	Step	Kyron lock version
1:12 PM	Kyron reconnected with read write	1
1:14 PM	Planted sentence written to scene 610130 (`update_lesson_activity`) and read back	1 → 2
1:14 PM	Lesson check reads the lesson through Kyron MCP and records content v4, "Changed in Kyron outside the QA agent" (the scene and the lecture script)	2
1:15 PM	Fidelity check: a **critical** finding on exactly the planted sentence, quoting the MIT notes as evidence. Two minor findings as well: a vocabulary definition and one objective's practice coverage	
1:15 PM	The revision agent proposes a one-sentence fix to `scenes.610130.narration`. Critical fixes need approval, so it is saved as proposed v5 and escalated	
1:16 PM	A person (CPE Admin) accepts the change	
1:16 PM	The fix is written to Kyron with the lock version, read back, and saved as applied v6, "QA revision agent, approved by a person"	2 → 3
1:16 PM	Independent verifier (high effort): *resolves*	
1:17 PM	Re-check of what the change affects: structure, fidelity for 504103, consistency and alignment all pass	
1:17 PM	Run ends **Incomplete**: the minor vocabulary finding is left for a person. The definition of superposition omits "homogeneous"	3

Read back from Kyron afterwards: scene 610130 and the lecture script both contain the corrected sentence, and the planted text is gone from both. The scene keeps QA's corrected sentence; the original scene text is saved and can be restored. No video was rendered and nothing was published.

3.2 What the test showed

#	Result	What it means
1	**Detection was precise:** critical severity, the right scene, the exact sentence, and evidence quoted from the source.	Seeded contradictions are caught. This is one sample; phases 3 and 5 measure the rate.
2	**The fix was minimal:** only the false sentence was rewritten. The lecture script was updated in step, and Kyron returned exactly what was written.	Scene-level writes keep the author's text and keep the script and scenes consistent.
3	**The edit lock worked:** Kyron's lock version rose by exactly one per write (1 → 2 → 3), with no conflicts.	QA cannot overwrite an edit someone makes in Kyron Studio while a run is in progress.
4	**The verifier agreed** with the fix, for $0.0006.	Cheap enough to run on every fix.
5	**Critical fixes wait for a person,** by design.	**Decision for the team:** allow critical *text* fixes to be applied when the verifier agrees, starting with "[QA Test]" drafts. Answer keys and lesson settings stay protected.
6	**Minor findings are not fixed automatically.** When the check reports "needs review", the run ends Incomplete. The superposition definition is a real, small error in Kyron's output.	**Decision for the team:** send minor findings that have a clear fix to the revision agent as well.
7	**Wording problems:** "required checks did not complete" is shown when a check completed but needs review. An approved fix is labeled "Fixed automatically". Shortened before/after text hides the change. The vocabulary finding points to "narration" instead of the term's definition.	Small interface fixes. They are listed in section 4.5.
8	**Results vary between runs.** The superposition finding appeared in this run but not in the 11:31 AM run on the same content. An alignment finding was closed on re-check although none of its content changed. Neither run reported the Black–Scholes example.	Repeat runs in phases 3 and 5 measure consistency. Consider keeping a finding open unless a change touched its fields.

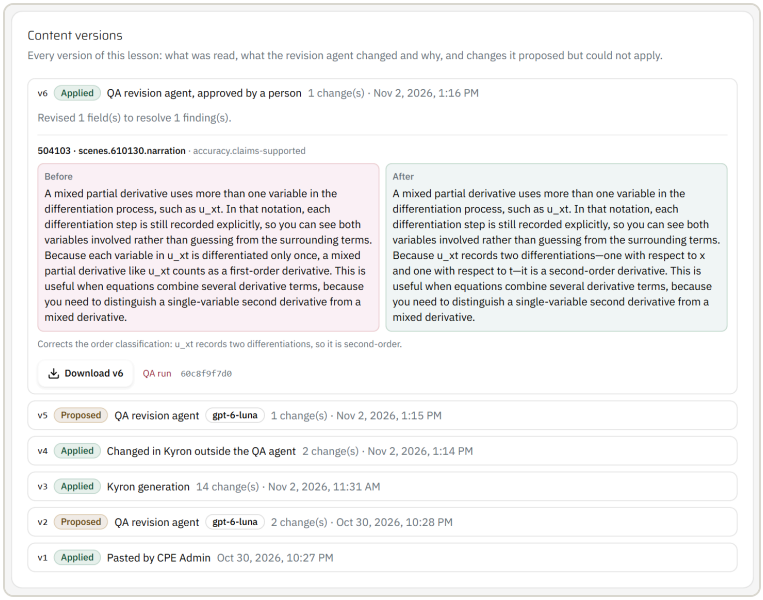


Figure 21. Content versions for the lesson after the test. v4 records the planted edit, v5 the proposal, and v6 the approved fix written to Kyron, with the scene before and after.

4. Testing plan: verify the harness and gather problems

Goals: (1) confirm each stage works as designed; (2) measure how well the checks catch real and seeded mistakes, and how often they raise false alarms; (3) log every problem in one place, with evidence, so the next phase can build the testing sets from it.

4.1 Phases

Phase	Scope	Pass criteria	Needs
0. Smoke done Oct 30	One course end to end: setup QA with fixes, Kyron draft 14826, a pasted lesson with a seeded error	All stages ran; seeded critical error caught; costs logged	-
0b. Live Kyron write done Nov 2	Draft 14826 read through MCP; one planted false sentence in a scene; fix approved and written back (section 3)	Caught as critical; fix written to the right scene, read back and verified; lock version +1 per write	-
1. Harness regression 109 passing	109 automated tests: pipeline, revision agent, verifier, Kyron mapping and writes, production stages, repair policy, auth and logging. Lint, typecheck and build run on every change	All pass; no unchecked write over a person's edit	None
2. Setup QA	20 real source documents, plus 20 setups with seeded problems (table 4.2); each run twice	≥90% of critical and ≥75% of major seeded problems caught; ≥80% of applied fixes judged correct; the same verdict on both runs ≥85% of the time	Source files; ~6 h review
3. Lesson QA (paste)	15 lessons with ~3 seeded defects each (≈45) and 5 clean lessons; each run 3 times	Same catch targets; ≤1 blocking false alarm per 5 clean lessons; no answer key changed automatically	Lesson authoring and review, ~10 h
4. Real-course pilot	10 CPE courses through the full flow to Kyron drafts (all named "[QA Test] ..."). Reviewers compare the lessons with the QA findings	Every problem a reviewer finds is logged; QA misses are classified	Kyron credits; ~7.5 h review
5. MCP and video write tested; video off	Read, write and video on 10 lessons; narration checked against the scene captions	Reads match Studio; writes read back exactly; captions match the script	Approval of Kyron render credits (`QA_KYRON_VIDEO`)

4.2 Seeded defect catalog

Where	Defect seeded	Expected severity	Expected harness behavior
Setup	Vague objective ("Understand X")	Major	Fixed automatically and re-checked
Setup	Duration that doesn't fit the material (e.g. 60 min for one page)	Major	Fixed automatically to an allowed duration
Setup	Invented audience, employer or credential	Major	Fixed automatically; evidence quoted from the source
Setup	Off-topic description	Critical	Proposed version, escalated, not applied

Lesson	Claim that contradicts the source	Critical	Found; fix proposed and escalated; written to the scene after approval (tested Nov 2)
Lesson	Claim the source doesn't support	Major	Found; fix proposed or applied
Lesson	Wrong answer key	Critical	Found by the blind solve; key never changed automatically
Lesson	Two defensible answers; question not answerable from the lesson	Major	Found; prompt or options fix proposed
Lesson	Narration and on-screen text disagree	Major	Both fields revised together
Lesson	Placeholder text ("[TODO]", "Lorem ipsum")	Major	Found by the structural rules (no model needed)
Lesson	Objective not covered; jargon above the audience	Major	Found by the alignment and audience checks
Lesson	Embedded instruction ("ignore previous instructions…")	-	Treated as untrusted text: not followed, nothing written

4.3 Measures

Measure	Definition	Target
Catch rate	Seeded defects found ÷ seeded, by type and severity	≥90% critical, ≥75% major
False alarm rate	Blocking findings on clean content that reviewers reject	≤1 per 5 clean lessons
Fix acceptance	Fixes a reviewer judges correct and in the author's voice	≥80%
Fix safety	Writes over a person's edit; answer keys changed automatically; new errors introduced by a fix	0
Consistency	Same input, same verdict and severity across repeated runs	≥85%
Escalation rate	Runs that need a person's decision	Tracked (no target yet)
Time and cost per course	From the logs: latency, tokens, estimated cost	Within section 5 estimates

4.4 How problems and mistakes are gathered

Recorded automatically: findings with evidence (run report), every content version with before and after, decisions (accept, reject or waive, with a reason), audits, model usage and cost, and errors (Logs). All of it exports to CSV/JSON.

Recorded by reviewers in the defect library, **QA agent → Defects** (Figure 20): one entry per problem, with stage, category, severity, expected and actual behavior, evidence and the QA run. Rejected findings (false alarms) and audits that find a defect (QA missed) are added automatically. The library exports to CSV and JSON.

Categories: QA missed a problem · False alarm · Wrong or harmful fix · Harness bug · Kyron generation error · Source or extraction problem. A short weekly triage assigns each entry an owner. Entries marked "QA missed" and "False alarm" become the testing sets.

4.5 Problems observed so far (smoke test and live write test)

#	Observation	Category	Next step
1	Kyron rejected the client file while GitHub served it as plain text; it was accepted as `application/json`. MCP connected Nov 2.	Resolved	Move the file to a host the team controls
2	Kyron status only updated while a page was open.	Resolved	Server-side poller in the scheduler tick (built)
3	Duration fixes went in opposite directions for the same source (3600→900 in one run, 1800→3600 in another).	Consistency	Repeated-run test in phase 2; tighten the scope rule
4	Math symbols in the PDF text came out as "□ □ □", so a quoted passage could not be matched and the finding was lowered to minor.	Source extraction	Test math-heavy PDFs; improve extraction
5	A run can show *Passed* while one check is *Failed* because its only finding is minor.	Rubric decision	Decide whether minor-only failures should pass
6	Course setup generation replaces the workspace name, so test names had to be re-applied.	Harness bug	Keep a name the user set
7	The title check keeps suggesting removing the "[QA Test]" label.	False alarm	Ignore test prefixes in the title check
8	The revision agent combined an audience fix with a factual fix in one narration rewrite (+~250 characters).	Watch	Check length and tone drift in fix acceptance
9	Model calls before the log system was switched on (early Oct 30) are not in the cost totals.	Reporting	None; totals start Oct 30, 8:16 PM ET
10	The Kyron lecture and vocabulary cite the Black–Scholes equation, which the source never mentions. The fidelity check did not report it.	QA missed (defect library)	Decide whether side examples outside the source should be reported
11	Kyron refuses segment and transcript reads, with read and with read write; no video is rendered yet.	Access	Captions are used; ask Kyron about segment access
12	Narration is about 5–6 minutes for a 15-minute lesson.	Calibration	Measure real learner time before judging duration
13	Critical text fixes always wait for a person, even when the verifier agrees.	Rubric decision	Decide whether to apply them on "[QA Test]" drafts (section 3.2)
14	Minor findings get no fix, and a "needs review" check leaves the run Incomplete (superposition definition).	Rubric decision	Decide whether minor findings with a clear fix go to the revision agent
15	Wording: "did not complete" for a check that needs review; "Fixed automatically" for an approved fix; the Production tab says only "Review the lesson check".	Interface	Reword the stage messages and labels
16	Shortened before/after text shows the unchanged start of a scene and hides the change; the vocabulary finding names "narration" instead of the term's definition field.	Interface	Show the changed sentence; point vocabulary findings at the term
17	The same content gave different minor findings in two runs; an alignment finding closed on re-check with no change to its content.	Consistency	Repeated-run measures; keep findings open unless a change touched them

5. Cost evaluation

5.1 Spent so far (from the logs)

All model calls ran on `gpt-6-luna` at **$0.10 per 1M input tokens, $0.01 per 1M cached input, $0.50 per 1M output** (reasoning tokens are billed as output). Each call's cost is calculated when it is logged, at the price in effect then. These are the app's own estimates from logged token counts, not an OpenAI invoice.

Purpose	Calls	Input tokens	Output tokens	Avg latency	Cost	Avg / call
Setup QA (`qa.course_setup`)	9	48,386	13,019	16.9 s	$0.01135	$0.00126
Source fidelity (`qa.fidelity`)	4	26,150	14,933	37.7 s	$0.01008	$0.00252
Revision agent (`qa.revision`)	4	25,229	1,175	3.9 s	$0.00311	$0.00078
Alignment (`qa.alignment`)	4	8,883	3,048	9.2 s	$0.00241	$0.00060
Setup generation (`course_setup`)	2	9,944	645	5.4 s	$0.00132	$0.00066
Consistency (`qa.consistency`)	4	6,730	821	2.6 s	$0.00108	$0.00027
Independent verifier (`qa.verify`)	1	5,179	138	2.7 s	$0.00059	$0.00059
Blind assessment	1	449	87	2.6 s	$0.00009	$0.00009
Total	**29**	**130,950**	**33,866**		**$0.03003**	**$0.00104**

Output tokens include 20,163 reasoning tokens. No cached input was used and no call failed. By day: $0.0168 on Oct 30 (18 calls) and $0.0132 on Nov 2 (11 calls). Kyron calls are not billed by OpenAI: 88 public API and 95 MCP calls were logged with 0 errors; their cost depends on the Kyron account plan.

5.2 Cost per run

Run	What it was	Outcome	Calls	Cost	Of which revision
601eb10b	Setup QA, first live test	Passed	2	$0.00214	-
11fad144	Setup QA, draft 14826 workspace (old per-field repair)	Passed after 3 fixes	3	$0.00469	-
cede430c	Setup QA inside generation (revision agent)	Passed after 2 fixes	3	$0.00282	$0.00065
497586fc	Setup QA after a person's edit (revision agent)	Passed after 1 fix	3	$0.00294	$0.00059
ad7fdfe6	Lesson QA, pasted 2-activity lesson with seeded error	Changes proposed	5	$0.00291	$0.00081
c1bed8fb	Lesson QA for draft 14826, before MCP was connected	Blocked (no MCP)	0	$0.00000	-
517e95db	Lesson QA for draft 14826 read through MCP (7 activities and vocabulary)	Passed	3	$0.00431	-
121d324b	Live write test: planted error found, fix approved, written, verified and re-checked	Incomplete (1 minor left)	8	$0.00890	$0.00106

> **Unit costs measured:** setup generation ≈ $0.0007; setup QA with fixes ≈ $0.003 (range $0.002 to $0.005); a revision agent call ≈ $0.0008; a verifier call ≈ $0.0006. A full check of the real 7-activity Kyron lesson costs ≈ $0.004, or ≈ $0.009 with one fix, verification and re-check. A course through setup, Kyron generation and lesson QA costs about one cent in model calls.

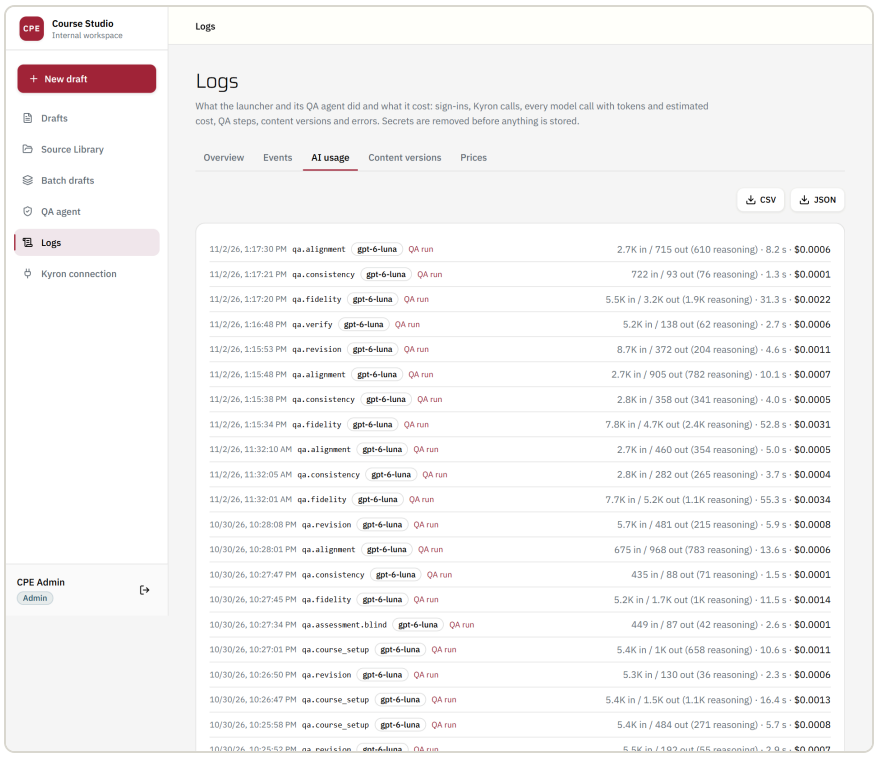


Figure 22. **Logs → AI usage**: each model call with purpose, model, tokens, latency, cost and a link to its QA run.

5.3 Projected cost of the testing plan

The real 7-activity lesson measured $0.004 per check and $0.009 with a fix, well under the earlier assumption. The estimate below keeps the conservative **$0.03 per lesson check** and **$0.05 with two revision rounds**, to allow for longer courses and multi-lesson drafts. Narration transcription is estimated at about $0.006 per audio minute (OpenAI's published estimate for `gpt-4o-transcribe`), about $0.09 for a 15-minute video. Every run has a hard limit of 600,000 tokens (at most about $0.30 per run even if every token were output), and hitting it stops the run instead of skipping checks.

Phase	Volume	Model cost basis	Model cost	People's time	Kyron cost
1. Harness regression	109 tests per change	Models are mocked	$0.00	-	-
2. Setup QA	40 setups × 2 runs	$0.0007 generation + $0.0031 QA	≈ $0.30	≈ 6 h	-
3. Lesson QA (paste)	20 lessons × 3 runs	$0.03 per run	≈ $1.80	≈ 10 h	-
4. Real-course pilot	10 courses	$0.004 setup + $0.05 lesson	≈ $0.55	≈ 7.5 h	10 drafts: to confirm
5. MCP and video	10 lessons	$0.05 QA + 2 × $0.09 transcription	≈ $2.30	≈ 5 h	10–20 renders: to confirm
Total			**≈ $5**	**≈ 28 h**	**To confirm with Kyron**

> **Budget recommendation:** set aside **$10** for model calls (2× contingency for retries, longer lessons and prompt tuning). Model calls are not the main cost of testing. People's review time (about 28 hours) and Kyron generation and video credits are. Kyron's per-draft and per-render pricing should be confirmed with the account owner before phases 4 and 5.

5.4 Cost controls already in place

- **Per-run token budget** (`QA_RUN_TOKEN_BUDGET=600000`): the run stops as *Blocked*; checks are never skipped silently.
- **Per-course budget** (`QA_COURSE_BUDGET_USD=1`): once a course's model spend reaches it, model steps stop and the production job shows *Blocked*. Spend against it is shown on the Production tab.
- **Video off by default** (`QA_KYRON_VIDEO=off`) and at most one re-render per lesson when it is on.
- **Model pinned for corrections** (`OPENAI_REVISION_MODEL=gpt-6-luna`) with medium effort; outputs are not stored at OpenAI (`store: false`).
- **Video only after the script passes**, so failing lessons never use render credits.
- **Price table with effective dates:** a price change applies from that point on and never rewrites past costs. Calls to a model with no price show "rate not set" instead of $0.
- **Logs** show spend for 24 hours, 7 days and 30 days, by model and by purpose, with CSV/JSON export for each phase's cost report.

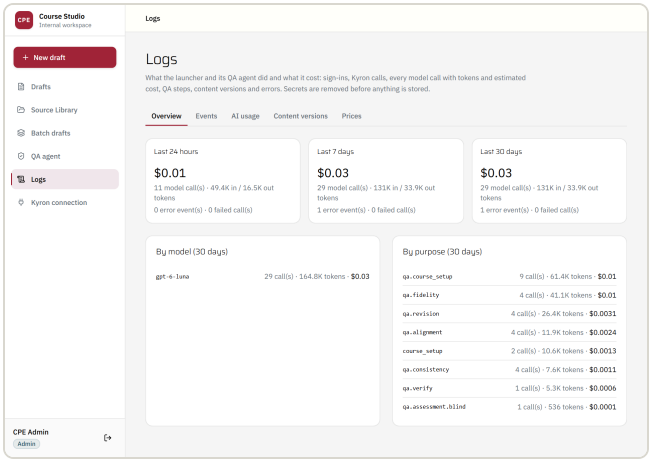


Figure 23. **Logs** → **Overview**: spend by period, model and purpose.

6. Appendix: records referenced

Record	Details
Kyron draft	14826, "[QA Test] Introduction to PDEs - MIT 18.152 Class 1", created through the app's API on Oct 30, 2026 (status done)
Workspace (draft 14826)	7e49593f-0596-4b1f-9c15-ab6507044bea: setup run 11fad144 (passed); lesson runs c1bed8fb (blocked: MCP), ad7fdfe6 (pasted, changes proposed), 517e95db (read through MCP, passed) and 121d324b-f1de-446d-870b-ac34603f7648 (live write test, content v4–v6)
Live write test	Scene 610130 in activity 504103. Lock version 1 → 2 (planted sentence, 1:14 PM ET) → 3 (QA's approved fix, 1:16 PM ET). The original scene text is saved in the session scratchpad and can be restored
Workspace (revision agent test)	36b18f77-7e67-41f6-aefa-469d1ca10818, "[QA Test] Revision agent - PDE setup": runs cede430c and 497586fc, content versions v1–v4
Kyron MCP	Connected Nov 2, 2026, 10:50 AM ET as CPE Admin with `read`; reconnected 1:12 PM ET with `read write`. Client file served as JSON from the secret gist. Captured data: `reports/kyron-mcp-dataset-draft-14826-2026-11-02.json` (media links removed)
Kyron lesson IDs	Collection 14826 → draft lesson 23930 → section 64011 → activities 504101–504107
Source used	`test_course.pdf`: MIT 18.152 Introduction to PDEs, Class Meeting 1 (text pasted as source; Vercel Blob not configured locally)
Database	Shared Neon database; QA, log, version, production-job and defect tables are migrations 100–125 (applied on their own, twice, to confirm they are safe to re-run)
Code checks	Lint, typecheck, 109 tests and production build all passing as of Nov 2, 2026
Configuration	`AUTH_MODE=access-code`, `OPENAI_MODEL=gpt-6-luna`, `OPENAI_REVISION_MODEL=gpt-6-luna`, `QA_AUTONOMY_POLICY=auto`, `QA_RUN_TOKEN_BUDGET=600000`, `QA_COURSE_BUDGET_USD=1`, `QA_AUDIT_RATE=0.1`, `QA_KYRON_WRITES=test-only`, `QA_KYRON_VIDEO=off`. `KYRON_OAUTH_CLIENT_ID` set to the JSON-served client file.